

ECommerce Project — Documentation

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Project for Advance Computer Programming

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1. Project Overview

An E-Commerce backend system built as an academic project demonstrating:

- **3 Design Patterns** (Factory, Strategy, Singleton)
- **5 Persistence Formats** (JSON, XML, Pickle, SQLite, GZIP + plain Text)
- **Full OOP** (ABC, Encapsulation, Polymorphism, Dataclass, Enums)
- **REST API** via Flask serving a live single-page dashboard

2. Directory Structure

```

ECommerce_Project/
├── app.py                Flask REST API server (entry point for web)
├── main.py              CLI demo / seed script (entry point for console)
├── requirements.txt      Python dependencies
├── models/              — Domain layer (WHAT the data is)
│   ├── product.py       Abstract base class + 3 concrete product types
│   ├── user.py          User dataclass + UserRole enum
│   └── order.py          Order class + OrderStatus enum
├── patterns/            — Design Patterns
│   ├── factory.py       ProductFactory (FACTORY pattern)
│   ├── singleton.py     Logger (SINGLETON pattern)
│   └── strategy.py      Payment methods (STRATEGY pattern)
├── persistence/         — Storage layer (5 formats)
│   ├── db_handler.py    SQLite (primary relational DB)
│   ├── json_handler.py  JSON (human-readable backup)
│   ├── xml_handler.py   XML (structured markup)
│   ├── pickle_handler.py Pickle (binary Python serialization)
│   └── gzip_handler.py  GZIP (compressed archive)
├── utils/
│   ├── helpers.py       buy_product, transfer, calculate_discount,
│   │                   save_transaction (→ data/transactions.txt)
│   └── exceptions.py    Custom exception hierarchy
├── tests/
│   └── test_suite.py     pytest unit tests
├── templates/
│   └── index.html        Single-page dashboard (Jinja2)
├── static/
│   ├── css/style.css    Dashboard styles
│   └── js/main.js        Frontend logic (API calls, modals, charts)
├── data/                — Auto-created at runtime
│   ├── database.db       SQLite database
│   ├── products.json     JSON backup
│   ├── products.xml      XML backup
│   ├── products.pkl      Pickle binary backup
│   ├── app.log           Singleton Logger output
│   ├── app_logs.gz       Compressed log archive
│   └── transactions.txt  Plain-text purchase ledger

```

3. Design Patterns

4.1 Factory Pattern — `patterns/factory.py`

Purpose: Create different Product types without exposing instantiation logic.

```

# Instead of this (bad – tight coupling):
if type == "electronics":
    p = Electronics(id, name, price, stock, warranty_years=2)
elif type == "clothing":
    p = Clothing(id, name, price, stock, size="M")

# We use this (Factory – loose coupling):
product = ProductFactory.create_product("electronics", 101, "Laptop", 95000, 13)

```

Input Type	Output Class	Extra Field
"electronics"	Electronics	warranty_years
"clothing"	Clothing	size
"grocery"	Grocery	expiry_date

3.2 Singleton Pattern — patterns/singleton.py

Purpose: Only ONE Logger instance exists across the entire application.

```
class Logger:
    _instance = None          # class-level variable – shared by ALL instances

    def __new__(cls):
        if cls._instance is None:
            cls._instance = super().__new__(cls)
            # open log file once here
        return cls._instance  # always returns the SAME object
```

Proof it works:

```
a = Logger()
b = Logger()
print(a is b)  # True – same object!
```

Why Singleton for Logger?

- Prevents multiple log files being opened simultaneously
- All modules write to the same data/app.log
- Avoids race conditions in file writing

3.3 Strategy Pattern — patterns/strategy.py

Purpose: Swap payment algorithms at runtime without if/elif chains.

```
# All payment classes implement the same interface (ABC):
class PaymentStrategy(ABC):
    @abstractmethod
    def pay(self, amount: float) -> bool: ...

class CreditCard(PaymentStrategy):
    def pay(self, amount): ...    # credit card logic

class JazzCash(PaymentStrategy):
    def pay(self, amount): ...    # jazzcash logic

class CashOnDelivery(PaymentStrategy):
    def pay(self, amount): ...    # COD logic
```

At runtime (in Order.checkout()):

```
# Order doesn't care WHICH payment – it just calls .pay()
self.payment.pay(self.total)  # polymorphism in action!
```

Strategy Class	Payment Method
CreditCard	Card number based
JazzCash	Mobile wallet

EasyPaiza Strategy Class	Mobile wallet Payment Method
CashOnDelivery	Physical cash

4. Persistence Layer

Why 5 formats?

Beacuae Lecture demands and :

Format	File	Why Used	Readable?
JSON	products.json	Primary backup, human-readable	☑ Yes
XML	products.xml	Used in enterprise web services	☑ Yes
Pickle	products.pkl	Full Python object serialization	☑ Binary
SQLite	database.db	Relational DB, SQL queries	Via tool
GZIP	app_logs.gz	Compressed log archive	After extract
Text	transactions.txt	Plain ledger (new.txt req. step 9)	☑ Yes

Write Strategy

```
Every Product CRUD operation:
1. Update PRODUCT_CATALOG dict (in-memory, instant)
2. save_product_db() → SQLite (durable)
3. save_products() → JSON (human-readable backup)

main.py also writes:
4. save_to_xml() → XML
5. save_to_pickle() → Pickle
6. save_transaction() → transactions.txt
7. Logger archives → app_logs.gz
```

Why is data saved TWICE (DB + JSON)?

Reason	Explanation
Redundancy	If one format corrupts, the other is still readable
Demo purpose	Academic requirement to show multiple persistence formats
Human readability	SQLite needs a viewer; JSON opens in any text editor
Flask API	Reads from SQLite (fast); JSON is a fallback export

5. OOP Concepts

5.1 Abstract Base Class (ABC) — `models/product.py`

```

from abc import ABC, abstractmethod

class Product(ABC):
    @abstractmethod
    def display_info(self) -> str: ...    # must be implemented by subclasses

    @abstractmethod
    def to_dict(self) -> dict: ...        # for serialization

```

Concrete subclasses: Electronics, Clothing, Grocery

5.2 Dataclass — `models/user.py`

```

from dataclasses import dataclass, field
from datetime import datetime

@dataclass
class User:
    name: str
    id: int
    email: str
    role: UserRole = UserRole.CUSTOMER
    created_at: str = field(default_factory=lambda: datetime.now().isoformat())

```

`@dataclass` auto-generates `__init__`, `__repr__`, `__eq__` — saves boilerplate.

5.3 Enum — `models/user.py`, `models/order.py`

```

from enum import Enum

class UserRole(Enum):
    CUSTOMER = "Customer"
    ADMIN    = "Admin"
    SELLER   = "Seller"

class OrderStatus(Enum):
    PENDING    = "Pending"
    CONFIRMED  = "Confirmed"
    SHIPPED    = "Shipped"
    DELIVERED  = "Delivered"
    CANCELLED  = "Cancelled"

```

Why Enum? Prevents typo bugs ("Custmer" vs "Customer"), enables IDE autocomplete, makes code self-documenting.

5.4 Encapsulation

```

class Product(ABC):
    def __init__(self, product_id, name, price, stock):
        self._product_id = product_id    # protected
        self._name        = name
        self.__price       = price        # private (name-mangled)
        self._stock        = stock

    @property
    def price(self):          # controlled read access
        return self.__price

    def update_stock(self, delta: int): # controlled write access
        self._stock += delta

```

5.5 Polymorphism

```
products = [Electronics(...), Clothing(...), Grocery(...)]
for p in products:
    print(p.display_info()) # calls DIFFERENT implementation for each type
```

6. API Endpoints

Products

Method	Endpoint	Description	Body / Query
GET	/api/products	List all products	?category=Electronics&in_stock=true
GET	/api/products/<id>	Single product	—
POST	/api/products	Create product	{type, product_id, name, price, stock}
PATCH	/api/products/<id>/stock	Update stock	{stock: 50}
DELETE	/api/products/<id>	Delete product	—

Users

Method	Endpoint	Description
GET	/api/users	All users
GET	/api/users/<id>	Single user
POST	/api/users	Create user
GET	/api/users/next-id	Auto-generate next ID

Orders

Method	Endpoint	Description
GET	/api/orders	All orders (?status=Confirmed)
GET	/api/orders/<id>	Single order
POST	/api/orders	Place new order
PATCH	/api/orders/<id>/status	Update status

System

Method	Endpoint	Description
GET	/api/stats	Dashboard statistics
GET	/api/logs?n=100	Last N log lines
GET	/api/datafiles	All data files with format/size
GET	/api/health	Health check

7. Requirements Coverage (new.txt)

Step	Requirement	Where Implemented
1	Products (3 types)	<code>models/product.py</code> + <code>patterns/factory.py</code>
2	Users with roles	<code>models/user.py</code> (dataclass + enum)
3	Orders	<code>models/order.py</code>
4	Payment strategies	<code>patterns/strategy.py</code>
5	Factory pattern	<code>patterns/factory.py</code>
6	Singleton Logger	<code>patterns/singleton.py</code>
7	Persistence (JSON/XML/Pickle)	<code>persistence/</code> folder
8	SQLite + GZIP	<code>persistence/db_handler.py</code> , <code>gzip_handler.py</code>
9	<code>transactions.txt</code>	<code>utils/helpers.py</code> → <code>save_transaction()</code>
10	<code>assert</code> + <code>pdb</code>	<code>utils/helpers.py</code> → <code>transfer()</code> , <code>calculate_discount()</code>
11	<code>tabulate</code> report	<code>main.py</code> Step 12 (grid-format product summary)
12	Flask REST API	<code>app.py</code> (14 endpoints)
13	Frontend Dashboard	<code>templates/index.html</code> + <code>static/js/main.js</code>